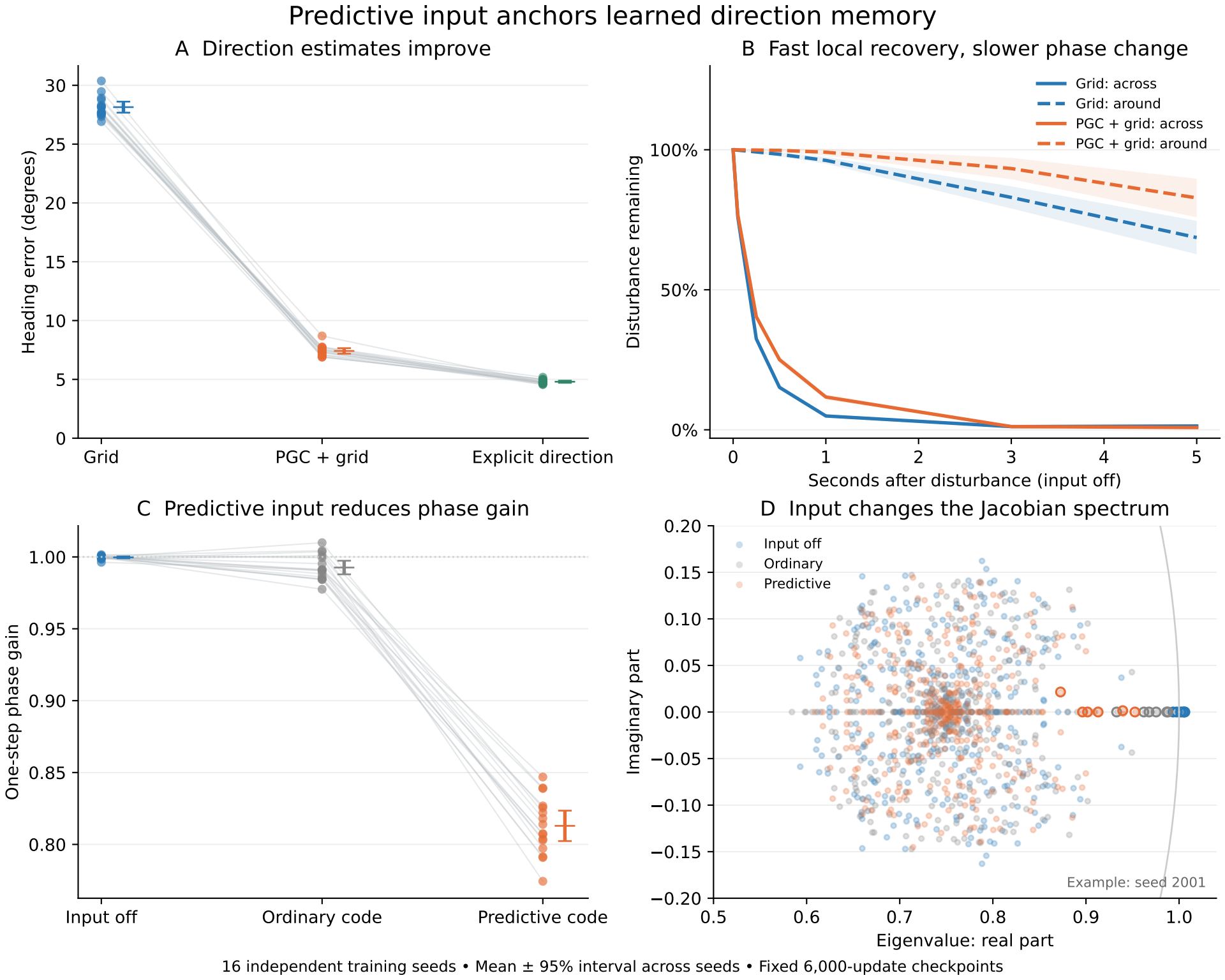

Predictive input anchors learned direction memory
A Direction estimates improve
Heading error (degrees)
30
25
20
15
10
5
0
Grid
PGC + grid
Explicit direction
B Fast local recovery, slower phase change
Grid: across
Grid: around
PGC + grid: across
PGC + grid: around
Disturbance remaining
100%
50%
0%
0
1
2
3
4
5
Seconds after disturbance (input off)
C Predictive input reduces phase gain
One-step phase gain
1.00
0.95
0.90
0.85
0.80
Input off
Ordinary code
Predictive code
D Input changes the Jacobian spectrum
Input off
Ordinary
Predictive
Imaginary part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
0.5
0.6
0.7
0.8
0.9
1.0
Eigenvalue: real part
Example: seed 2001
16 independent training seeds • Mean ± 95% interval across seeds • Fixed 6,000-update checkpoints